

EDUCATION	Purdue University <i>Ph.D. Candidate in Mathematics</i> 2021 – 2027 (expected) • Research Interests: Inverse Problems & Partial Differential Equations • Advisors: Peijun Li & Isaac Harris
	New Jersey Institute of Technology <i>M.S. in Applied Mathematics</i> May 2021 • Advisor: Christina Frederick
	Rutgers University <i>B.A. in Mathematics</i> May 2018
PUBLICATIONS	1. I. Harris, P. Li, G. Ozochiawaeze. Sampling methods for the inverse cavity scattering problem of biharmonic waves. <i>Submitted to Inverse Problems</i>, 2025. arXiv:2509.02773 2. I. Harris, P. Li, and G. Ozochiawaeze. Factorization method for the biharmonic scattering problem for an absorbing penetrable obstacle. <i>Under Preparation</i>.
WORK EXPERIENCE	MIT Lincoln Laboratory Lexington, MA May 2023 – Oct 2023 • <i>Summer Research Intern, Group 36 – Integrated Missile Defense Technology</i> • Applied compressive sensing to enhance radar imaging for missile target detection, improving reconstruction accuracy. • Presented compelling research findings on compressed radar sensing, demonstrating potential applications in enhanced defense technologies.
	MIT Lincoln Laboratory Lexington, MA May 2022 – Aug 2022 • <i>Summer Research Intern, Group 37 – Advanced Undersea Systems & Technology</i> • Developed algorithms to analyze coherence loss in sonar signals, improving underwater detection and tracking. • Applied sonar array interferometry to extract spatial information from acoustic wave-fronts.
TEACHING	• Lecturer, Purdue University — <i>MA 16020: Applied Calculus II</i>, Fall 2025 • Recitation Instructor, Purdue University — <i>MA 261: Multivariable Calculus</i>, Spring 2024 • Teaching Assistant, Purdue University — <i>MA 511: Linear Algebra and Applications</i>, Spring 2023 • Recitation Instructor, Purdue University — <i>MA 261: Multivariable Calculus</i>, Fall 2022 • Recitation Instructor, Purdue University — <i>MA 162: Calculus II</i>, Spring 2022 • Recitation Instructor, Purdue University — <i>MA 161: Calculus I</i>, Fall 2021 • Instructor, Accel Learning — Mentored students in preparation for AMC and AIME math competitions, Spring–Fall 2020 • Math Instructor, Mathnasium, K-12 Teaching & Tutoring, Fall 2018–Spring 2019

AWARDS AND HONORS	• SIAM (Society of Industrial & Applied Mathematics) Student Travel Award	2025
	• NSF Computational Mathematics Program Award, DMS-2208256,	2023-2024
	• GEM Fellowship	2022
	• Graduate C/Startup Research Grant	2020
	• Tau Sigma Honors Society	2016

SKILLS **Computing & Software:** Python, C++, MATLAB, COMSOL, FEniCS

CONFERENCES ATTENDED	Summer School: Theory and Applications of Elliptic PDE (Presented)	2025 UC Irvine
	Mathematics and Statistics in Industry Workshop and Panel Discussion	2025 Purdue University
	Conference on Mathematics of Wave Phenomena	2025 Karlsruhe Institute of Technology (KIT)
	Paris-Saclay Conference in Analysis and PDE	2024 Laboratoire de Mathématiques d'Orsay
	SIAM Student Chapter Conference (Presented)	2024 Purdue University
	Workshop on Scientific Computing	2022 Purdue University
	New Ideas in Computational Inverse Problems	2022 BIRS, Banff International Research Station

SEMINAR & CONFERENCE TALKS	Extending Qualitative Reconstruction to Biharmonic Scattering with Limited Data	2025
	<i>Summer School: Theory and Applications of Elliptic PDE, UC Irvine</i>	
	Extending Qualitative Reconstruction in Biharmonic Scattering from Limited Data	2025
	<i>Graduate Student Analysis Seminar, Purdue University</i>	
	On Sampling Methods for Recovering a Clamped Cavity in a Thin Plate	2025
	<i>CCAM Lunch Seminar, Purdue University</i>	
	A Linear Sampling Method for Recovering a Clamped Cavity in a Thin Plate	2024
	<i>Purdue Graduate Research Day, Purdue University</i>	
	A Linear Sampling Method for Recovering a Clamped Cavity in a Thin Plate	2024
	<i>SIAM Student Chapter Conference, Purdue University</i>	
	Comparing and Integrating the Probe Method and Method of Singular Sources	2023
	<i>Graduate Student Analysis Seminar, Purdue University</i>	
	The Factorization Method in Inverse Scattering	2023
	<i>Graduate Student Analysis Seminar, Purdue University</i>	
	Compressive Radar Imaging in Missile Defense	2023
	<i>Integrated Missile Defense Technology Group, MIT Lincoln Laboratory</i>	
	Quantifying Real Operational Performances of SONAR Arrays in Random Ocean environments	2022
	<i>Advanced Undersea Systems & Technology Group, MIT Lincoln Laboratory</i>	
	Finite Element Modeling of Underwater Acoustic Environments	2022
	<i>Purdue Math Student Colloquium, Purdue University</i>	

INFORMAL TALKS	The Dawn of the Diophantine Era: Algebra at the Crossroads of Empires <i>Math History Seminar, Purdue University</i>	2025
	Fixed Points in Action: from Brouwer to Lefschetz & Beyond <i>Topolodays, Purdue University</i>	2025
	The Life, Mathematics, and Political Advocacy of Sofya Kovalevskaya <i>Math History Seminar, Purdue University</i>	2024
	The K-theoretic Atiyah-Singer Index Theorem <i>Student Operator Algebras Seminar, Purdue University</i>	2024
	Introducing Pseudodifferential Operators <i>Purdue Math Student Colloquium</i>	2024
	Bourbaki and the Foundations of Modern Math <i>Math History Seminar, Purdue University</i>	2023
	Wasan: Mathematical Treasures in Early Modern Japan <i>Math History Seminar, Purdue University</i>	2022
SERVICE	Purdue Math Department Graduate Representative <i>Purdue University</i>	2025-2026
	Mentor, Directed Reading Project in Computational Topology <i>Purdue University — mentee: Aaron (Sang Hyun) Kim</i>	2025
	Founder and Organizer, Math History Seminar <i>Purdue University</i>	2022
	Rutgers Undergraduate Mathematics Association Treasurer & Web Manager <i>Rutgers University</i>	2017-2018